

# YIBO LIN

Associate Professor ◇ Department of Design Automation and Computing System

School of Integrated Circuits, 5 Yiheyuan Road, Beijing, China, 100871 ◇ Peking University

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## RESEARCH INTERESTS

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Modeling and optimization in VLSI CAD, machine learning applications, and heterogeneous computing

## EDUCATION

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**University of Texas at Austin, Texas, USA** Aug. 2013 – May 2018

Ph.D., Department of Electrical and Computer Engineering

**Shanghai Jiao Tong University, Shanghai, China** Sep. 2009 – Jun. 2013

B.S., Department of Microelectronics

## EXPERIENCE

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**Peking University, Beijing, China** Aug. 2025 – present

Associate Professor

**Peking University, Beijing, China** Jul. 2019 – Jul. 2025

Assistant Professor

Department of Design Automation and Computing System,

School of Integrated Circuits (since Nov. 2021)

Center for Energy-efficient Computing and Applications (CECA),

School of Electronics Engineering and Computer Science

**University of Texas at Austin, Texas, U.S.** Jun. 2018 – Jun. 2019

Postdoc

Department of Electrical and Computer Engineering

## TEACHING EXPERIENCE

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Practice in AI-Assisted VLSI Circuit Design

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| Instructor | Graduate | 2026 |
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Fundamental Algorithms for Engineering in Integrated Circuits

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| Instructor | Graduate | 2023–2026 |
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Design Automation and Computing System

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| Instructor | Graduate | 2022–2024 |
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Optimization and Machine Learning in VLSI Design Automation

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| Instructor | Undergraduate | 2021–2024 |
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Introduction to Computing B

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| Instructor | Undergraduate | 2020–2022 |
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## AWARDS AND HONORS

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|---------------------------------------------------------|-------------------------------|------|
| Best Paper Award Nomination (13/596)                    | DAC                           | 2026 |
| Best Paper Award Nomination (×3) (12/145)               | ISEDA                         | 2026 |
| Best Paper Award Nomination (19/266)                    | ICCAD                         | 2025 |
| Best Paper Award Nomination (7/420)                     | DAC                           | 2025 |
| Honorable Mention (×2) (/135)                           | ISEDA                         | 2025 |
| Best Paper Award Nomination (13/168)                    | ASPDAC                        | 2025 |
| Best Paper Award Nomination (10/195)                    | ICCAD                         | 2024 |
| Best Paper Award (6/151) & Honorable Mention            | ISEDA                         | 2024 |
| Early Career Award (only one per year)                  | CCF Technical Committee in IC | 2023 |
| Inaugural Best Reviewer Award                           | ICCAD                         | 2023 |
| Best Paper Award (4/205)                                | DATE                          | 2023 |
| Best Paper Award (4/249)                                | DATE                          | 2022 |
| Best Paper Award Nomination (6/132)                     | ICCAD                         | 2022 |
| Donald O. Pederson Best Paper Award (2/3495 in 4 years) | TCAD                          | 2021 |
| Best Paper Award                                        | ISPD                          | 2020 |
| Best Paper Award Nomination (12/86)                     | ASPDAC                        | 2020 |
| Best Paper Award (1/201) & Nomination (5/201)           | DAC                           | 2019 |
| Best Paper Award Nomination                             | ISPD                          | 2019 |
| Inaugural Best Paper Award                              | Integration, the VLSI Journal | 2018 |
| Franco Cerrina Memorial Best Student Paper Award        | SPIE                          | 2016 |
| A. Richard Newton Young Student Fellow                  | DAC                           | 2014 |

## PROFESSIONAL SERVICE

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### Executive Committee Member

- General co-chair, ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD) 2025–2026
- Program co-chair, ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD) 2024
- Panel chair, IEEE International Symposium of Electronics Design Automation (ISEDA) 2023–2024
- Financial chair, ACM/IEEE Workshop on Machine Learning for CAD (MLCAD) 2021–2023

### Journal Editor

- Associate Editor, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024 – present
- Associate Editor, ACM Transaction on Design Automation of Electronic Systems (TODAES), 2024 – present
- Associate Editor, Elsevier Integration, the VLSI Journal (Integration), 2024 – present
- Guest Editor, ACM Transaction on Design Automation of Electronic Systems (TODAES) Special Issue on MLCAD, 2024
- Guest Editor, ACM Transaction on Design Automation of Electronic Systems (TODAES) Special Issue on MLCAD, 2022

### Contest Chair

- Contest Problem Chair, Interated Circuits EDA Elite Challenge 2021 – 2023

#### Technical Program Committee Member

- ACM/IEEE Design Automation Conference (DAC): 2020
- IEEE/ACM International Conference on Computer-Aided Design (ICCAD): 2018 – 2021, 2023
- Design, Automation and Test in Europe Conference (DATE): 2025
- IEEE International Conference on Computer Design (ICCD): 2019
- IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC): 2021 – 2022
- ACM International Symposium on Physical Design (ISPD): 2020, 2025
- ACM/IEEE Workshop on Machine Learning for CAD (MLCAD): 2021
- ACM Great Lakes Symposium on VLSI (GLVLSI): 2024
- Workshop on Synthesis And System Integration of Mixed Information technologies (SASIMI): 2021
- IEEE Electron Devices Technology and Manufacturing Conference (EDTM): 2021
- IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS): 2022.

#### Journal Reviewer

- IEEE Transaction on Computer-Aided Design of Integrated Circuits and Systems (TCAD)
- IEEE Transactions on Computers (TC)
- ACM Transaction on Design Automation of Electronic Systems (TODAES)
- SPIE Journal of Micro/Nanolithography, MEMS, and MOEMS (JM3)
- Elsevier, Integration, the VLSI Journal (Integration)

#### Volunteer

- ACM SIGDA Website Administrator, 2021 – present

## 10 SELECTED PUBLICATIONS

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Inverse chronological order. \* denotes corresponding authors.

1. Yifan Chen, Zaiwen Wen, Yun Liang and **Yibo Lin\***, “[Stronger Mixed-Size Placement Backbone Considering Second-Order Information](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Francisco, CA, Oct 29-31, 2023.
2. Haoyi Zhang, Xiaohan Gao, Haoyang Luo, Jiahao Song, Xiyuan Tang, Junhua Liu, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[SAGERoute: Synergistic Analog Routing Considering Geometric and Electrical Constraints with Manual Design Compatibility](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Apr 17-19, 2023. (**Best Paper Award**)
3. Jing Mai, Jiarui Wang, Zhixiong Di and **Yibo Lin\***, “[Multi-Electrostatic FPGA Placement Considering SLICEL-SLICEM Heterogeneity, Clock Feasibility, and Timing Optimization](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Sep, 2023.
4. Zizheng Guo, Tsung-Wei Huang and **Yibo Lin\***, “[Accelerating Static Timing Analysis using CPU-GPU Heterogeneous Parallelism](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2023.

5. Qipan Wang, Xiaohan Gao, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[DeePEB: A Neural Partial Differential Equation Solver for Post Exposure Baking Simulation in Lithography](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Diego, CA, Nov 01-03, 2022. (**Best Paper Nomination**)
6. Zizheng Guo and **Yibo Lin\***, “[Differentiable-Timing-Driven Global Placement](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.
7. Siting Liu, Peiyu Liao, Zhitang Chen, Wenlong Lv, **Yibo Lin\*** and Bei Yu\*, “[FastGR: Global Routing on CPU-GPU with Heterogeneous Task Graph Scheduler](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Mar 14-23, 2022. (**Best Paper Award**)
8. Zizheng Guo, Jing Mai and **Yibo Lin\***, “[Ultrafast CPU/GPU Kernels for Density Accumulation in Placement](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.
9. Zizheng Guo, Tsung-Wei Huang and **Yibo Lin\***, “[A Provably Good and Practically Efficient Algorithm for Common Path Pessimism Removal in Large Designs](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.
10. **Yibo Lin\***, Zixuan Jiang, Jiaqi Gu, Wuxi Li, Shounak Dhar, Haoxing Ren, Brucek Khailany and David Z. Pan, “[DREAMPlace: Deep Learning Toolkit-Enabled GPU Acceleration for Modern VLSI Placement](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Apr, 2021. (**Best Paper Award**)

## PUBLICATIONS

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### Book Chapters

- [B3] **Yibo Lin**, Zizheng Guo and Jing Mai, “[Deep Learning Framework for Placement](#)”, Machine Learning Applications in Electronic Design Automation, Springer, 2023, edited by Haoxing Ren and Jiang Hu. (**Invited Book Chapter**)
- [B2] Haoyu Yang, **Yibo Lin** and Bei Yu, “[Machine Learning for Mask Synthesis and Verification](#)”, Machine Learning Applications in Electronic Design Automation, Springer, 2023, edited by Haoxing Ren and Jiang Hu. (**Invited Book Chapter**)

### Conference and Journal Papers (\* denotes corresponding authors)

*Summary: DAC (37), ICCAD (24), IEEE TCAD (33), DATE (21), etc.*

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- [J240] Zhenkun Lin, **Yibo Lin**, Genggeng Liu\* and Gang Du, “[URoute: Universal Routability Prediction](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)
- [J239] Xu He\*, Yushan Wang, Chenjing Yang, Renjun Zhao, **Yibo Lin**, Peiyu Liao, Bei Yu, Yao Wang, Chang Liu and Yang Guo, “[Efficient Timing Prediction and Optimization Using Derivable Gradient Boosting Machine Model at Placement Stage](#)”, ACM Transactions on Design Automation of Electronic Systems (TODAES), 2026. (accepted)
- [J238] Chunyuan Zhao, Zizheng Guo, Zuodong Zhang and **Yibo Lin\***, “[GAP-LA: GPU-Accelerated Performance-Driven Layer Assignment](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

- [J237] Xufeng Yao, Jiayi Jiang, Yuxuan Zhao, Peiyu Liao, **Yibo Lin** and Bei Yu\*, “EvoPlace: Evolution of Optimization Algorithms for Global Placement via Large Language Models”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)
- [C236] Zixuan Pan, Zizheng Guo, Yufan Du, Runsheng Wang and **Yibo Lin**\*, “Disentangled Differentiable Timing-Power Co-Optimization with Quad-Gradient Gate sizing”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (**Best Paper Nomination**)
- [C235] Yi Ren, Yukun Wang, Xiang Meng, Guoyao Cheng, Baokang Peng, Lining Zhang, **Yibo Lin**, Runsheng Wang and Guangyu Sun\*, “Celle: Automated Standard Cell Library Extension via Equality Saturation”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.
- [C234] Jing Mai, Yi-Chen Lu, **Yibo Lin** and Haoxing Ren, “LATTE: Legality-Assured Differentiable Timing-Driven Detailed Placement”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.
- [C233] Wenbo An, Kairong Guo, Haoyi Zhang and **Yibo Lin**\*, “EPiCell: Electro-Physical Co-Modeling for Standard Cell PPA Prediction”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.
- [C232] Yuhao Ji, Yunqi Shi, Tianshu Hou, Yuxuan Zhao, Chunyuan Zhao, Peiyu Liao, Zizheng Guo, Chao Qian, **Yibo Lin** and Bei Yu\*, “Integrated Timing-driven Placement for Hybrid-Bonding-based Face-to-Face 3D ICs”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.
- [C231] Kexing Zhou, Youwei Zhuo, **Yibo Lin**, Weikang Qian, Pengpeng Ren and Yun Liang\*, “Activity-Aware Partitioning for Effective Multi-Threaded Event-Driven RTL Simulation”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.
- [C230] Jiechen Huang, Hector R. Rodriguez, Dingcheng Yang, Zuochang Ye, **Yibo Lin** and Wenjian Yu\*, “AttentionCap: Transformer Based Capacitance Matrix Learning Toward Full-Chip Extraction”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.
- [C229] Haoyi Zhang, Weijian Fan, Xiaohan Gao, Bingyang Liu, Runsheng Wang and **Yibo Lin**\*, “PANDA: An LLM-Enhanced Performance-Driven Analog Design Framework Bridging Design Intent and Layout Generation”, ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (**Invited Paper**)
- [C228] Haoyi Zhang, Kairong Guo, Bojie Zhang, **Yibo Lin**\* and Runsheng Wang, “[FusionCell: Cross-Attentive Fusion of Layout Geometry and Netlist Topology for Standard-Cell Performance Prediction](#)”, International Conference on Machine Learning (ICML), Seoul, Korea, Jul 6-11, 2026.
- [C227] Xizhe Shi, Zizheng Guo, Zuodong Zhang, Yun Liang, Runsheng Wang\* and **Yibo Lin**\*, “HeteroPower: A CPU-GPU Heterogeneous Engine to Accelerate Gate-Level Power Analysis”, IEEE/ACM International Symposium of EDA (ISED), Singapore, May 8-12, 2026. (**Best Paper Nomination**)
- [C226] Kairong Guo, Jiechen Huang, Wenjian Yu and **Yibo Lin**\*, “CAPCell: Standard Cell Layout

- Synthesis with Parasitic Capacitance Aware Parallel Sampling”, IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.
- [C225] Jincheng Lou, Ruohan Xu, Jiecheng Ma, Runzhe Tao, Xinyu Qu and **Yibo Lin\***, “LEGO: An LLM Skill-Based Front-End Design Generation Platform”, IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026. (**Best Paper Nomination**)
- [C224] Zizhuo Fu, Yifan Zhou, Zhaoxin Lu, Guangyu Sun, Runsheng Wang, Meng Li\* and **Yibo Lin\***, “RePart: Efficient Hypergraph Partitioning with Logic Replication Optimization for Multi-FPGA System”, IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.
- [C223] Xun Jiang, Haoran Lu, Yifan Chen, Jiarui Wang, Chunyuan Zhao, Yibo Zhang, Jianxiang Jin, Heng Wu, Runsheng Wang and **Yibo Lin\***, “Flexible Double-Side Pin Redistribution with Efficient Virtual-Net-Guided Cell Type Assignment”, IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026. (**Best Paper Nomination**)
- [C222] Haichuan Liu, Zizheng Guo, Runsheng Wang and **Yibo Lin\***, “ElmoreCeff: A GPU-Friendly Elmore-Like Delay Calculator with a Closed-Form Effective Capacitance Model”, IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.
- [C221] Bingyang Liu, Haoning Jiang, Haoyi Zhang, Xiaohan Gao, Zichen Kong, Xiyuan Tang, David Z. Pan and **Yibo Lin\***, “GRAIN: A Design-Intent-Driven Analog Layout Migration Framework”, IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.
- [C220] Yuxiang Zhao, Jing Mai, Zishu Li, Haoyi Zhang, Jincheng Lou, Qing He, Runsheng Wang and **Yibo Lin\***, “PowerCube: A Formula-in-the-Loop Framework for Cross-Stage Power Map Prediction”, IEEE/ACM International Symposium of EDA (ISEDA), Singapore, May 8-12, 2026.
- [C219] Mingjun Wang, Yihan Wen, Yuntao Lu, Fengrui Liu, Yuxiang Zhao, Boyu Han, Jianan Mu, **Yibo Lin**, Runsheng Wang, Huawei Li and Bei Yu, “[CircuitNet 3.0: A Multi-Modal Dataset with Task-Oriented Augmentation for AI-Driven Circuit Design](#)”, International Conference on Learning Representations (ICLR), Rio de Janeiro, Brazil, Apr 23-27, 2026.
- [C218] Chenglin Ye, Yuze Lu, Yizhan Liu, Ligong Zhang, Jinghan Xu, Fei Liu, **Yibo Lin**, Zheng Zhou and Xiaoyan Liu, “EMAPER: Cross-Level Electromigration Aware Placement and Routing EDA Workflow for Interconnects Hotspot Prediction and Mitigation”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.
- [C217] Jiajun Tan, Qichao Ma, Yiming Du, Yiming Gan, Ling Liang, **Yibo Lin**, Ming Zhu, Zongwei Wang and Yimao Cai, “SONIC: Smart Optimization for Neural-Integrated CMP with Timing-Aware Fills”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.
- [C216] Zizheng Guo, Runsheng Wang and **Yibo Lin\***, “Provably Optimal Planar Pareto Nearest Neighbor Search with Double Monotone Chains”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.
- [C215] Jianqing Chen, Zhenkun Lin, Xun Jiang, Genggeng Liu\*, **Yibo Lin** and Gang Du, “DSR: A Systematic Approach for Efficient Double-Sided Signal Routing”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.

- [C214] Siting Liu, Peng Xu, Peiyu Liao, Keren Zhu, **Yibo Lin** and Bei Yu\*, “Submodular Maximization-Inspired Adaptive Routing Bend Space Planning”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.
- [C213] Tianxiang Zhu, Qipan Wang, **Yibo Lin**\* and Runsheng Wang\*, “Efficient Warpage Simulation Of Complex 2.5-D/3-D IC Structures with Novel Meshing Algorithm and Layerwise Plate Theory”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Verona, Italy, Apr 20-22, 2026.
- [C212] Yuhao Ji, Zizheng Guo, **Yibo Lin** and Bei Yu\*, “[HeteroSTA3D: A CPU-GPU Heterogeneous Static Timing Analysis Engine for 3D ICs](#)”, China Semiconductor Technology International Conference (CSTIC), Shanghai, Mar, 2026. (**Invited Paper**)
- [C211] Jiarui Wang, Runzhe Tao, Jing Mai, Xun Jiang, Shenghua Wang, Cuiliu Yang, Haoyu Jie, Kan Huang, Richard Y. Sun and **Yibo Lin**\*, “TDM Signal Grouping and Package Pin Assignment for 2.5D Multi-FPGA Systems with Lookahead Placement”, International Symposium on Field-Programmable Gate Arrays (FPGA), Monterey, CA, Feb 22-24, 2026.
- [C210] Zizheng Guo, Haichuan Liu, Xizhe Shi, Shenglu Hua, Zuodong Zhang, Chunyuan Zhao, Runsheng Wang and **Yibo Lin**\*, “HeteroSTA: A CPU-GPU Heterogeneous Static Timing Analysis Engine with Holistic Industrial Design Support”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026. (**Invited Paper**)
- [C209] Xizhe Shi, Zizheng Guo, **Yibo Lin**\*, Zuodong Zhang, Yun Liang and Runsheng Wang\*, “HeteroLatch: A CPU-GPU Heterogeneous Latch-Aware Timing Analysis Engine”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.
- [C208] Kairong Guo, Haoran Lu, Rui Guo, Jiarui Wang, Chunyuan Zhao, Heng Wu, Runsheng Wang and **Yibo Lin**\*, “Standard Cell Layout Synthesis for Dual-Sided 3D-Stacked Transistors”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.
- [C207] Weijian Fan, Haoyi Zhang, Weibin Lin, Runsheng Wang and **Yibo Lin**\*, “MOSTAR: Multi-Stage Hierarchical Bayesian Optimization for Substructure-Aware High-Dimensional Analog Circuit Sizing”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.
- [C206] Chenhao Xue, Kezhi Li, Jiaxing Zhang, Yi Ren, Zhengyuan Shi, Chen Zhang, **Yibo Lin**, Lining Zhang, Qiang Xu and Guangyu Sun\*, “AC-Refiner: Efficient Arithmetic Circuit Optimization Using Conditional Diffusion Models”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.
- [C205] Jingchao Hu, **Yibo Lin**, Hao Yu, Quan Chen, Zhou Jin\* and Cheng Zhuo, “IR Drop-Aware ECO: A Fast Approach to Minimize Layout and Timing Disturbance”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Hong Kong, Jan, 2026.
- [J204] Siting Liu, Ziyi Wang, Fangzhou Liu, **Yibo Lin**, Bei Yu\* and Martin Wong, “[Sign-off Timing Considerations via Concurrent Routing Topology Optimization](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)
- [J203] Zhenkun Lin, Genggeng Liu\*, Xing Huang, **Yibo Lin**, Jixin Zhang, Wenhao Liu and Ting-Chi Wang, “[A Unified Deep Reinforcement Learning Approach for Constructing Rectilinear](#)



- and [Octilinear Steiner Minimum Tree](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)
- [J202] Jingchen Zhu, Chenhao Xue, Yiqi Chen, Zhao Wang, Chen Zhang, Yu Shen, Yifan Chen, Zekang Cheng, Yu Jiang, Tianqi Wang, **Yibo Lin**, Wei Hu, Bin Cui, Runsheng Wang, Yun Liang and Guangyu Sun\*, “[Theseus: Exploring Efficient Wafer-Scale Chip Design for Large Language Models](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)
- [J201] Yajuan Su, Zixi Liu, **Yibo Lin**, Xiaojing Su, Yuqin Wang, Xin Hong, Yujie Jiang, Pengyu Ren and Yayi Wei, “[A Post-Routing Layout Optimization Framework for Lithography Process Window Enlargement](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)
- [J200] Jing Mai, Chunyuan Zhao, Zuodong Zhang, Zhixiong Di, Runsheng Wang and **Yibo Lin**\*, “[LEGALM 2.0: A Versatile Augmented Lagrangian Method-Based Methodology for Mixed-Cell-Height Legalization](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2025. (accepted)
- [C199] Chunyuan Zhao, Jiarui Wang, Xun Jiang, Jincheng Lou and **Yibo Lin**\*, “GTA: GPU-Accelerated Track Assignment with Lightweight Lookup Table for Conflict Detection”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025. (**Best Paper Nomination**)
- [C198] Yufan Du, Zizheng Guo, Runsheng Wang and **Yibo Lin**\*, “Differentiable Physical Optimization”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [C197] Yuhao Ji, Yuntao Lu, Zuodong Zhang, Zizheng Guo, **Yibo Lin** and Bei Yu\*, “DiffCCD: Differentiable Concurrent Clock and Data Optimization”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [C196] Yuan Pu, Yuhao Ji, Siying Yu, Zuodong Zhang, Zizheng Guo, Zhuolun He, **Yibo Lin**, David Z. Pan and Bei Yu\*, “GPU Acceleration for Versatile Buffer Insertion”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [C195] Yi Ren, Baokang Peng, Chenhao Xue, Kairong Guo, Yukun Wang, Guoyao Cheng, **Yibo Lin**, Lining Zhang and Guangyu Sun\*, “Orthrux: Dual-Loop Automated Framework for System-Technology Co-Optimization”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [C194] Haichuan Liu, Zizheng Guo, Runsheng Wang and **Yibo Lin**\*, “IncreGPUSTA: GPU-Accelerated Incremental Static Timing Analysis for Iterative Design Flows”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [C193] Tianxiang Zhu, Qipan Wang, **Yibo Lin**\* and Runsheng Wang\*, “High-Resolution Full-Chip Thermal Resistance Extraction of BEOL Interconnects in 3-D ICs Considering Detailed Via Connectivity”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [C192] Chenchen Zhao, Zhengyuan Shi, Xiangyu Wen, Chengjie Liu, Yi Liu, Yunhao Zhou, Yuxiang Zhao, Hefei Feng, Yinan Zhu, Waa Gwok-Wan, Xin Cheng, Weiyu Chen, Yongqi Fu, Chujie Chen, Chenhao Xue, Ying Wang, **Yibo Lin**, Jun Yang, Ning Xu, Xi Wang and Qiang Xu\*,



- “MMCCircuitEval: A Comprehensive Multimodal Circuit-Focused Benchmark for Evaluating LLMs”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Munich, Germany, Oct, 2025.
- [J191] Bingyang Liu, Haoyi Zhang, Xiaohan Gao, Zichen Kong, Xiyuan Tang, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[LayoutCopilot: An LLM-powered Multi-agent Collaborative Framework for Interactive Analog Layout Design](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Aug, 2025.
- [C190] Yuxiang Zhao, Zhuomin Chai, Xun Jiang, Qiang Xu, Runsheng Wang and **Yibo Lin\***, “[DeepLayout: Learning Neural Representations of Circuit Placement Layout](#)”, International Conference on Machine Learning (ICML), Vancouver, Canada, Jul 13-19, 2025.
- [C189] Lijie Zeng, Jiatai Sun, Xiao Wu, Dan Niu, Tianshi Wang, **Yibo Lin**, Zuochang Ye and Zhou Jin\*, “[G-SpNN: GPU-Accelerated Passivity Enforcement for S-Parameter Modeling with Neural Networks](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.
- [C188] Ziyang Yu, Peng Xu, Zixiao Wang, Binwu Zhu, Qipan Wang, **Yibo Lin**, Runsheng Wang, Bei Yu\* and Martin Wong, “[SDM-PEB: Spatial-Depthwise Mamba for Enhanced Post-Exposure Bake Simulation](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.
- [C187] Zizheng Guo, Yanqing Zhang, Runsheng Wang, **Yibo Lin** and Haoxing Ren, “[GEM: GPU-Accelerated Emulator-Inspired RTL Simulation](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025. (**Best Paper Nomination**)
- [C186] Xun Jiang, Haoran Lu, Yuxuan Zhao, Jiarui Wang, Zizheng Guo, Heng Wu, Bei Yu, Sung Kyu Lim, Runsheng Wang, Ru Huang and **Yibo Lin\***, “[A Systematic Approach for Multi-Objective Double-Side Clock Tree Synthesis](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.
- [C185] Jiarui Wang, Yanjing Liu and **Yibo Lin\***, “[Synergistic Die-Level Router for Multi-FPGA System with Time-Division Multiplexing Optimization](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.
- [C184] Yifan Chen, Jing Mai, Zuodong Zhang and **Yibo Lin\***, “[RUPlace: Optimizing Routability via Unified Placement and Routing Formulation](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 22-25, 2025.
- [J183] Yuxiang Zhao, Zhuomin Chai, Xun Jiang, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[PDNNet: PDN-Aware GNN-CNN Heterogeneous Network for Dynamic IR Drop Prediction](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2025.
- [J182] Rui Guo, Haoran Lu, Jiacheng Sun, Xun Jiang, Lining Zhangm, Ming Li, **Yibo Lin**, Runsheng Wang, Heng Wu\* and Ru Huang, “[Design Optimization of Flip FET Standard Cells With Dual-Sided Pins for Ultimate Scaling](#)”, IEEE Transactions on Electron Devices (TED), Jun, 2025.
- [C181] Haoyi Zhang, Shizhao Sun, **Yibo Lin**, Runsheng Wang and Jiang Bian, “[AnalogXpert: Automating Analog Topology Synthesis by Incorporating Circuit Design Expertise into Large](#)

- Language Models”, IEEE/ACM International Symposium of EDA (ISED), Hong Kong, May 9-12, 2025.
- [C180] Yufan Du, Zizheng Guo, Yang Hsu, Zhili Xiong, Seunggeun Kim, David Z. Pan, Runsheng Wang and **Yibo Lin\***, “[Addressing Continuity and Expressivity Limitations in Differentiable Physical Optimization: A Case Study in Gate Sizing](#)”, IEEE/ACM International Symposium of EDA (ISED), Hong Kong, May 9-12, 2025. (**Honorable Mention Paper Award**)
- [C179] Kairong Guo and **Yibo Lin\***, “[Multi-Row Standard Cell Layout Synthesis with Enhanced Scalability](#)”, IEEE/ACM International Symposium of EDA (ISED), Hong Kong, May 9-12, 2025.
- [C178] Qipan Wang, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[ATSim3.5D: A Multiscale Thermal Simulator for 3.5D-IC Systems based on Nonlinear Multigrid Method](#)”, IEEE/ACM International Symposium of EDA (ISED), Hong Kong, May 9-12, 2025. (**Honorable Mention Paper Award**)
- [C177] Chenhao Xue, Yi Ren, Jinwei Zhou, Kezhi Li, Chen Zhang, **Yibo Lin**, Lining Zhang, Qiang Xu and Guangyu Sun\*, “[DOMAC: Differentiable Optimization for High-Speed Multipliers and Multiply-Accumulators](#)”, IEEE/ACM International Symposium of EDA (ISED), Hong Kong, May 9-12, 2025.
- [C176] Yi Ren, Chenhao Xue, Jiaying Zhang, Chen Zhang, Qiang Xu, **Yibo Lin** and Guangyu Sun\*, “[DiffuSE: Cross-Layer Design Space Exploration of DNN Accelerator via Diffusion-Driven Optimization](#)”, IEEE/ACM International Symposium of EDA (ISED), Hong Kong, May 9-12, 2025.
- [C175] Haoran Lu, Xun Jiang, Yanbang Chu, Ziqiao Xu, Rui Guo, Wanyue Peng, **Yibo Lin**, Runsheng Wang, Heng Wu\* and Ru Huang, “[A Tale of Two Sides of Wafer: Physical Implementation and Block-Level PPA on Flip FET with Dual-sided Signals](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Lyon, France, Mar 31, 2025.
- [C174] Tianxiang Zhu, Qipan Wang, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[MORE-Stress: Model Order Reduction based Efficient Numerical Algorithm for Thermal Stress Simulation of TSV Arrays in 2.5D/3D IC](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Lyon, France, Mar 31, 2025.
- [C173] Xizhe Shi, Zizheng Guo, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[Handling Latch Loops in Timing Analysis with Improved Complexity and Divergent Loop Detection](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Lyon, France, Mar 31, 2025.
- [C172] Haikang Diao, Haoyi Zhang, Jiahao Song, Haoyang Luo, **Yibo Lin**, Runsheng Wang, Yuan Wang and Xiyuan Tang\*, “[SEGA-DCIM: Design Space Exploration-Guided Automatic Digital CIM Compiler with Multiple Precision Support](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Lyon, France, Mar 31, 2025.
- [C171] Jing Mai, Chunyuan Zhao, Zuodong Zhang, Zhixiong Di, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[LEGALM: Efficient Legalization for Mixed-Cell-Height Circuits with Linearized Augmented Lagrangian Method](#)”, ACM International Symposium on Physical Design (ISPD), Austin, TX, Mar 16-19, 2025.
- [J170] Xun Jiang, Jiarui Wang, Jing Mai, Zhixiong Di and **Yibo Lin\***, “[A Robust FPGA Router With Optimization of High-Fanout Nets and Intra-CLB Connections](#)”, IEEE Transactions on

- [J169] Yuxuan Zhao, Peiyu Liao, Siting Liu, Jiayi Jiang, **Yibo Lin** and Bei Yu\*, “[Analytical Heterogeneous Die-to-Die 3D Placement With Macros](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Feb, 2025.
- [C168] Che Chang, Boyang Zhang, Cheng-Hsiang Chiu, Dian-Lun Lin, Yi-Hua Chung, Wan-Luan Lee, Zizheng Guo, **Yibo Lin** and Tsung-Wei Huang\*, “[PathGen: An Efficient Parallel Critical Path Generation Algorithm](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan, 2025. (**Best Paper Nomination**)
- [C167] Boyang Zhang, Che Chang, Cheng-Hsiang Chiu, Dian-Lun Lin, Yang Sui, Chih-Chun Chang, Yi-Hua Chung, Wan Luan Lee, Zizheng Guo, **Yibo Lin** and Tsung-Wei Huang\*, “[iTAP: An Incremental Task Graph Partitioner for Task-parallel Static Timing Analysis](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan, 2025.
- [C166] Zizheng Guo, **Yibo Lin**\*, Runsheng Wang and Ru Huang, “[Analyzing Timing in Shorter Time: A Journey through Heterogeneous Parallelism for Static Timing Analysis](#)”, IEEE International Conference on Solid-State and Integrated Circuit Technology (ICSICT), Zhuhai, China, Oct 22-25, 2024. (**Invited Paper**)
- [J165] Wenshuo Yue, Teng Zhang, Zhaokun Jing, Kai Wu, Yuxiang Yang, Zhen Yang, Yongqin Wu, Weihai Bu, Kai Zheng, Jin Kang, **Yibo Lin**, Yaoyu Tao, Bonan Yan\*, Ru Huang and Yuchao Yang\*, “[A scalable universal Ising machine based on interaction-centric storage and compute-in-memory](#)”, Nature Electronics, Oct, 2024.
- [C164] Qipan Wang, Xueqing Li, Tianyu Jia, **Yibo Lin**\*, Runsheng Wang and Ru Huang, “[AT-Place2.5D: Analytical Thermal-Aware Chiplet Placement Framework for Large-Scale 2.5D-IC](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024.
- [C163] Chunyuan Zhao, Zizheng Guo, Rui Wang, Zaiwen Wen, Yun Liang and **Yibo Lin**\*, “[HeLEM-GR: Heterogeneous Global Routing with Linearized Exponential Multiplier Method](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024.
- [C162] Zizheng Guo, Zuodong Zhang, Wuxi Li, Tsung-Wei Huang, Xizhe Shi, Yufan Du, **Yibo Lin**\*, Runsheng Wang and Ru Huang, “[HeteroExcept: A CPU-GPU Heterogeneous Algorithm to Accelerate Exception-aware Static Timing Analysis](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024.
- [C161] Xiaohan Gao, Haoyi Zhang, Bingyang Liu, **Yibo Lin**\*, Runsheng Wang and Ru Huang, “[Joint Placement Optimization for Hierarchical Analog/Mixed-Signal Circuits](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024.
- [C160] Yufan Du, Zizheng Guo, **Yibo Lin**\*, Runsheng Wang and Ru Huang, “[Fusion of Global Placement and Gate Sizing with Differentiable Optimization](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024. (**Best Paper Nomination**)
- [C159] Tianxiang Zhu, Qipan Wang, **Yibo Lin**\*, Runsheng Wang and Ru Huang, “[FaStTherm: Fast and Stable Full-Chip Transient Thermal Predictor Considering Nonlinear Effects](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024.

- [C158] Jing Mai, Zuodong Zhang, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[MORPH: More Robust ASIC Placement for Hybrid Region Constraint Management](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), New Jersey, Oct, 2024.
- [J157] Tsung-Yi Ho, Sadaf Khan, Jinwei Liu, Yi Liu, Zhengyuan Shi, Ziyi Wang, Qiang Xu\*, Evangeline F.Y. Young, Bei Yu, Ziyang Zheng, Binwu Zhu, Keren Zhu, Yiqi Che, Yun Liang, **Yibo Lin**, Guojie Luo, Guangyu Sun, Runsheng Wang, Xinming Wei, Chenhao Xue, Haoyi Zhang, Zuodong Zhang, Yuxiang Zhao, Sunan Zou, Lei Chen, Yu Huang, Min Li, Dimitrios Tsaras, Mingxuan Yuan, Hui-Ling Zhen, Zhufei Chu, Wenji Fang, Xingquan Li and Zhiyao Xie, “[Large Circuit Models: Opportunities and Challenges](#)”, Science China Information Sciences, Sep, 2024.
- [C156] Jiarui Wang, Xun Jiang and **Yibo Lin\***, “[Top-Level Routing for Multiply-Instantiated Blocks with Topology Hashing](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.
- [C155] Yufan Du, Zizheng Guo, Xun Jiang, Zhuomin Chai, Yuxiang Zhao, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[PowPrediCT: Cross-Stage Power Prediction with Circuit-Transformation-Aware Learning](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.
- [C154] Haoyi Zhang, Jiahao Song, Xiaohan Gao, Xiyuan Tang, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[EasyACIM: An End-to-End Automated Analog CIM with Synthesizable Architecture and Agile Design Space Exploration](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.
- [C153] Zichen Kong, Xiyuan Tang\*, Wei Shi, Yiheng Du, **Yibo Lin** and Yuan Wang, “[PVT sizing: A TuRBO-RL-Based Batch-Sampling Optimization Framework for PVT-Robust Analog Circuit Synthesis](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.
- [C152] Yuan Pu, Fangzhou Liu, Yu Zhang, Zhuolun He, Kai-Yuan Chao, **Yibo Lin** and Bei Yu\*, “[Lesyn: Placement-aware Logic Resynthesis for Non-Integer Multiple-Cell-Height Designs](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.
- [C151] Wan Luan Lee, Dian-Lun Lin, Tsung-Wei Huang\*, Shui Jiang, Tsung-Yi Ho, **Yibo Lin** and Bei Yu, “[G-kway: Multilevel GPU-Accelerated k-way Graph Partitioner](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.
- [C150] Chenhao Xue, Chen Zhang, Xun Jiang, Gao Zhutianya, **Yibo Lin** and Guangyu Sun\*, “[Oltron: Algorithm-Hardware Co-design for Outlier-Aware Quantization of LLMs with Inter-/Intra-Layer Adaptation](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 23-27, 2024.
- [C149] Haoran Lu, Yandong Ge, Xun Jiang, Jiacheng Sun, Wanyue Peng, Rui Guo, Ming Li, **Yibo Lin**, Runsheng Wang, Heng Wu\* and Ru Huang, “[First Experimental Demonstration of Self-Aligned Flip FET \(FFET\): A Breakthrough Stacked Transistor Technology with 2.5T Design, Dual-Side Active and Interconnects](#)”, IEEE Symposium on VLSI Technology and Circuits (VLSI), Honolulu, HI, Jun 16-20, 2024.
- [J148] Peiyu Liao, Yuxuan Zhao, Dawei Guo, **Yibo Lin** and Bei Yu\*, “[Analytical Die-to-Die 3D Placement With Bistratal Wirelength Model and GPU Acceleration](#)”, IEEE Transactions on

- [C147] Xiaohan Gao, Haoyi Zhang, Zhu Pan, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[Migrating Standard Cells for Multiple Drive Strengths by Routing Imitation](#)”, IEEE/ACM International Symposium of EDA (ISEDA), Xi'an, China, May 10-13, 2024.
- [C146] Qipan Wang, Tianxiang Zhu, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[ATSim3D: Towards Accurate Thermal Simulator for Heterogeneous 3D IC Systems Considering Nonlinear Leakage and Conductivity](#)”, IEEE/ACM International Symposium of EDA (ISEDA), Xi'an, China, May 10-13, 2024. (**Honorable Mention Paper Award**)
- [C145] Jing Mai, Jiarui Wang, Yifan Chen, Zizheng Guo, Xun Jiang, Yun Liang and **Yibo Lin\***, “[OpenPARF 3.0: Robust Multi-Electrostatics Based FPGA Macro Placement Considering Cascaded Macros Groups and Fence Regions](#)”, IEEE/ACM International Symposium of EDA (ISEDA), Xi'an, China, May 10-13, 2024. (**Best Paper Award**)
- [C144] Xun Jiang, Zhuomin Chai, Yuxiang Zhao, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[CircuitNet 2.0: An Advanced Dataset for Promoting Machine Learning Innovations in Realistic Chip Design Environment](#)”, International Conference on Learning Representations (ICLR), Vienna, Austria, May 7-11, 2024.
- [J143] Yufei Chen, Zizheng Guo, Runsheng Wang, Ru Huang, **Yibo Lin** and Cheng Zhuo, “[Dynamic Supply Noise Aware Timing Analysis With JIT Machine Learning Integration](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), May, 2024. (accepted)
- [C142] Zizheng Guo, Tsung-Wei Huang, Zhou Jin, Cheng Zhuo, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[Heterogeneous Static Timing Analysis with Advanced Delay Calculator](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Valencia, Spain, Mar 24-28, 2024.
- [C141] Haoyi Zhang, Xiaohan Gao, Zilong Shen, Jiahao Song, Xiaoxu Cheng, Xiyuan Tang, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[SAGERoute 2.0: Hierarchical Analog and Mixed Signal Routing Considering Versatile Routing Scenarios](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Valencia, Spain, Mar 24-28, 2024.
- [C140] Yuan Pu, Tinghuan Chen, Zhuolun He, Chen Bai, Haisheng Zheng, **Yibo Lin** and Bei Yu\*, “[IncreMacro: Incremental Macro Placement Refinement](#)”, ACM International Symposium on Physical Design (ISPD), Taipei, Mar 12-15, 2024. (**Best Paper Nomination**)
- [C139] Yu Zhang, Yuan Pu, Fangzhou Liu, Peiyu Liao, Kaiyuan Chao, Keren Zhu, **Yibo Lin** and Bei Yu\*, “[Multi-Electrostatics Based Placement for Non-Integer Multiple-Height Cells](#)”, ACM International Symposium on Physical Design (ISPD), Taipei, Mar 12-15, 2024.
- [C138] Siting Liu, Jiayi Jiang, Zhuolun He, Ziyi Wang, **Yibo Lin** and Bei Yu\*, “[Routing-aware Legal Hybrid Bonding Terminal Assignment for 3D Face-to-Face Stacked ICs](#)”, ACM International Symposium on Physical Design (ISPD), Taipei, Mar 12-15, 2024.
- [J137] Zhixiong Di\*, Runzhe Tao, Jing Mai, Lin Chen and **Yibo Lin**, “[LEAPS: Topological-Layout-Adaptable Multi-Die FPGA Placement for Super Long Line Minimization](#)”, IEEE Transactions on Circuits and Systems I, Mar, 2024.
- [J136] Jing Mai, Jiarui Wang, Zhixiong Di and **Yibo Lin\***, “[Multielectrostatic FPGA Placement Considering SLICEL-SLICEM Heterogeneity, Clock Feasibility, and Timing Optimization](#)”,



IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Feb, 2024.

- [C135] Cheng-Hsiang Chiu, Zhicheng Xiong, Zizheng Guo, Tsung-Wei Huang\* and **Yibo Lin**, “[An Efficient Task-parallel Pipeline Programming Framework](#)”, International Conference on High-Performance Computing in Asia-Pacific Region (HPC Asia), Nagoya, Japan, Jan, 2024.
- [J134] Zizheng Guo, Tsung-Wei Huang and **Yibo Lin**\*, “[Accelerating Static Timing Analysis using CPU-GPU Heterogeneous Parallelism](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Dec, 2023.
- [J133] Zhuomin Chai, Yuxiang Zhao, Wei Liu\*, **Yibo Lin**\*, Runsheng Wang and Ru Huang, “[CircuitNet: An Open-Source Dataset for Machine Learning in VLSI CAD Applications with Improved Domain-Specific Evaluation Metric and Learning Strategies](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Dec, 2023.
- [J132] Jing Mai, Jiarui Wang, Zhixiong Di and **Yibo Lin**\*, “[OpenPARF: 基于深度学习工具包的大规模异构 FPGA 开源布局布线框架](#)”, 电子与信息学报, 2023.
- [J131] Zuodong Zhang, Zizheng Guo, **Yibo Lin**\*, Meng Li\*, Runsheng Wang and Ru Huang, “[AVATAR: An Aging- and Variation-Aware Dynamic Timing Analyzer for Error-Efficient Computing](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Nov, 2023.
- [J130] Guannan Guo, Tsung-Wei Huang\*, **Yibo Lin**, Zizheng Guo, Sushma Yellapragada and Martin Wong, “[A GPU-accelerated Framework for Path-based Timing Analysis](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Nov, 2023.
- [C129] Yifan Chen, Zaiwen Wen, Yun Liang and **Yibo Lin**\*, “[Stronger Mixed-Size Placement Backbone Considering Second-Order Information](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Francisco, CA, Oct 29-31, 2023.
- [J128] Peiyu Liao, Dawei Guo, Zizheng Guo, Siting Liu, Zhitang Chen, Wenlong Lv, **Yibo Lin**\* and Bei Yu\*, “[DREAMPlace 4.0: Timing-driven Placement with Momentum-based Net Weighting and Lagrangian-based Refinement](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Oct, 2023.
- [J127] Xiaohan Gao, Haoyi Zhang, Siyuan Ye, Mingjie Liu, David Z. Pan, Linxiao Shen, Runsheng Wang, **Yibo Lin**\* and Ru Huang, “[Post-Layout Simulation Driven Analog Circuit Sizing](#)”, SCIENCE CHINA Information Sciences, Oct, 2023.
- [C126] Jing Mai, Jiarui Wang, Zhixiong Di, Guojie Luo, Yun Liang and **Yibo Lin**\*, “[OpenPARF: An Open-Source Placement and Routing Framework for Large-Scale Heterogeneous FPGAs with Deep Learning Toolkit](#)”, International Conference on ASIC (ASICON), Nanjing, China, Oct, 2023. (**Invited Paper**)
- [C125] Xun Jiang, Zizheng Guo, Zhuomin Chai, Yuxiang Zhao, **Yibo Lin**\*, Runsheng Wang and Ru Huang, “[Accelerating Routability and Timing Optimization with Open-Source AI4EDA Dataset CircuitNet and Heterogeneous Platforms](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Francisco, CA, Oct, 2023. (**Invited Paper**)
- [C124] Kexing Zhou, Yun Liang\*, **Yibo Lin**, Runsheng Wang and Ru Huang, “[Khronos: Fusing Memory Access for Improved Hardware RTL Simulation](#)”, IEEE/ACM International Symposium on Microarchitecture (MICRO), Toronto, Canada, Oct, 2023.

- [J123] Binwu Zhu, Xinyun Zhang, **Yibo Lin**, Bei Yu and Martin Wong, “[DRC-SG 2.0: Efficient Design Rule Checking Script Generation via Key Information Extraction](#)”, ACM Transactions on Design Automation of Electronic Systems (TODAES), Sep, 2023.
- [C122] Zizheng Guo, Zuodong Zhang, Xun Jiang, Wuxi Li, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[General-Purpose Gate-Level Simulation with Partition-Agnostic Parallelism](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 9-13, 2023.
- [C121] Qipan Wang, Ping Liu, Ligguo Jiang, Mingjie Liu, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[MTL-Designer: An Integrated Flow for Analysis and Synthesis of Microstrip Transmission Line](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 9-13, 2023.
- [C120] Peiyu Liao, Hongduo Liu, **Yibo Lin\***, Bei Yu\* and Martin Wong, “[On a Moreau Envelope Wirelength Model for Analytical Global Placement](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 9-13, 2023.
- [C119] Siting Liu, Ziyi Wang, Fangzhou Liu, **Yibo Lin**, Bei Yu and Martin Wong\*, “[Concurrent Sign-off Timing Optimization via Deep Steiner Points Refinement](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 9-13, 2023.
- [C118] Su Zheng, Lancheng Zou, Siting Liu, **Yibo Lin**, Bei Yu and Martin Wong\*, “[Mitigating Distribution Shift for Congestion Optimization in Global Placement](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 9-13, 2023.
- [C117] Yu Zhang, Yifan Chen, Zhonglin Xie, Hong Xu, Zaiwen Wen, **Yibo Lin\*** and Bei Yu\*, “[LRSDP: Low-Rank SDP for Triple Patterning Lithography Layout Decomposition](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 9-13, 2023.
- [J116] Siting Liu, Yuan Pu, Peiyu Liao, Hongzhong Wu, Rui Zhang, Zhitang Chen, Wenlong Lv, **Yibo Lin\*** and Bei Yu\*, “[FastGR: Global Routing on CPU-GPU with Heterogeneous Task Graph Scheduler](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jul, 2023.
- [C115] Yuxiang Zhao, Zhuomin Chai, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[HybridNet: Dual-Branch Fusion of Geometrical and Topological Views for VLSI Congestion Prediction](#)”, IEEE/ACM International Symposium of EDA (ISEDA), Nanjing, China, May 8-11, 2023.
- [C114] Haoyi Zhang, Xiaohan Gao, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[Multi-Scenario Analog and Mixed-Signal Circuit Routing with Agile Human Interaction](#)”, IEEE/ACM International Symposium of EDA (ISEDA), Nanjing, China, May 8-11, 2023.
- [C113] Haoyi Zhang, Xiaohan Gao, Haoyang Luo, Jiahao Song, Xiyuan Tang, Junhua Liu, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[SAGERoute: Synergistic Analog Routing Considering Geometric and Electrical Constraints with Manual Design Compatibility](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Apr 17-19, 2023. (**Best Paper Award**)
- [C112] Zuodong Zhang, Meng Li\*, **Yibo Lin**, Runsheng Wang and Ru Huang, “[READ: Reliability-Enhanced Accelerator Dataflow Optimization using Critical Input Pattern Reduction](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Apr 17-19, 2023.



- [J111] Xiaohan Gao, Haoyi Zhang, Mingjie Liu, Linxiao Shen, David Z. Pan, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[Interactive Analog Layout Editing with Instant Placement and Routing Legalization](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Mar, 2023.
- [J110] **Yibo Lin**, Avi Ziv and Haoxing Ren, “[Introduction to the Special Issue on Machine Learning for CAD/EDA](#)”, ACM Transactions on Design Automation of Electronic Systems (TODAES), Mar, 2023.
- [J109] Xinfu Zhang, Zuodong Zhang, **Yibo Lin\***, Zhigang Ji\*, Runsheng Wang and Ru Huang, “[Efficient Aging-Aware Standard Cell Library Characterization Based on Sensitivity Analysis](#)”, IEEE Transactions on Circuits and Systems II: Express Briefs, Feb, 2023.
- [C108] Yifan Chen, Jing Mai, Xiaohan Gao, Muhan Zhang and **Yibo Lin\***, “[MacroRank: Ranking Macro Placement Solutions Leveraging Translation Equivariance](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 16-19, 2023.
- [C107] Jiarui Wang, Jing Mai, Zhixiong Di and **Yibo Lin\***, “[A Robust FPGA Router with Concurrent Intra-CLB Rerouting](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 16-19, 2023.
- [J106] Junchi Yan\*, Xianglong Lyu, Ruoyu Cheng and **Yibo Lin**, “Towards Machine Learning for Placement and Routing in Chip Design: a Methodological Overview”, arXiv preprint, 2022.
- [J105] Wei Li, Jialu Xia, Yuzhe Ma, Jialu Li, **Yibo Lin** and Bei Yu\*, “[Adaptive Layout Decomposition with Graph Embedding Neural Networks](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Nov, 2022.
- [C104] Zizheng Guo, Feng Gu and **Yibo Lin\***, “[GPU-Accelerated Rectilinear Steiner Tree Generation](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Diego, CA, Nov 01-03, 2022.
- [C103] Qipan Wang, Xiaohan Gao, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[DeePEB: A Neural Partial Differential Equation Solver for Post Exposure Baking Simulation in Lithography](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Diego, CA, Nov 01-03, 2022. (**Best Paper Nomination**)
- [C102] **Yibo Lin\***, Xiaohan Gao, Haoyi Zhang, Runsheng Wang and Ru Huang, “[Intelligent and Interactive Analog Layout Design Automation](#)”, IEEE International Conference on Solid-State and Integrated Circuit Technology (ICSICT), Nanjing, China, Oct 25-28, 2022. (**Invited Paper**)
- [J101] Zizheng Guo, Mingwei Yang, Tsung-Wei Huang and **Yibo Lin\***, “[A Provably Good and Practically Efficient Algorithm for Common Path Pessimism Removal in Large Designs](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Oct, 2022.
- [J100] Martin Rapp, Hussam Amrouch, **Yibo Lin**, Bei Yu, David Z. Pan, Marilyn Wolf and Jörg Henkel\*, “[MLCAD: A Survey of Research in Machine Learning for CAD](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Oct, 2022. (**Invited Keynote paper**)
- [C99] Binwu Zhu, Xinyun Zhang, **Yibo Lin**, Bei Yu\* and Martin Wong, “[Efficient Design Rule](#)

- Checking Script Generation via Key Information Extraction”, ACM/IEEE Workshop on Machine Learning for CAD (MLCAD), Snowbird, Utah, Sep 12-13, 2022.
- [J98] **Yibo Lin**, Tong Qu, Zongqing Lu, Yajuan Su\* and Yayi Wei\*, “[Asynchronous Reinforcement Learning Framework and Knowledge Transfer for Net Order Exploration in Detailed Routing](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Sep, 2022.
- [J97] Zhuomin Chai, Yuxiang Zhao, **Yibo Lin\***, Wei Liu, Runsheng Wang and Ru Huang, “[CircuitNet: An Open-Source Dataset for Machine Learning Applications in Electronic Design Automation \(EDA\)](#)”, SCIENCE CHINA Information Sciences, Sep, 2022.
- [C96] Jing Mai, Yibai Meng, Zhixiong Di and **Yibo Lin\***, “[Multi-Electrostatic FPGA Placement Considering SLICEL-SLICEM Heterogeneity and Clock Feasibility](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.
- [C95] Zizheng Guo and **Yibo Lin\***, “[Differentiable-Timing-Driven Global Placement](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.
- [C94] Zizheng Guo, Mingjie Liu, Jiaqi Gu, Shuhan Zhang, David Z. Pan and **Yibo Lin\***, “[A Timing Engine Inspired Graph Neural Network Model for Pre-Routing Slack Prediction](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.
- [C93] Zuodong Zhang, Zizheng Guo, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[AVATAR: An Aging- and Variation-Aware Dynamic Timing Analyzer for Application-based DVAFS](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.
- [C92] Bowen Wang, Guibao Shen, Dong Li, Jianye Hao, Wulong Liu, Yu Huang, Hongzhong Wu, **Yibo Lin**, Guangyong Chen and Pheng Ann Heng, “[LHNN: Lattice Hypergraph Neural Network for VLSI Congestion Prediction](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.
- [J91] Tsung-Wei Huang\*, Dian-Lun Lin, Chun-Xun Lin and **Yibo Lin**, “[Taskflow: A Lightweight Parallel and Heterogeneous Task Graph Computing System](#)”, IEEE Transactions on Parallel and Distributed Systems (TPDS), Jun, 2022.
- [C90] Zuodong Zhang, Zizheng Guo, **Yibo Lin\***, Runsheng Wang and Ru Huang, “[EventTimer: Fast and Accurate Event-Based Dynamic Timing Analysis](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Mar 14-23, 2022.
- [C89] Siting Liu, Peiyu Liao, Zhitang Chen, Wenlong Lv, **Yibo Lin\*** and Bei Yu\*, “[FastGR: Global Routing on CPU-GPU with Heterogeneous Task Graph Scheduler](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Mar 14-23, 2022. (**Best Paper Award**)
- [C88] Peiyu Liao, Siting Liu, Zhitang Chen, Wenlong Lv, **Yibo Lin\*** and Bei Yu\*, “[DREAMPlace 4.0: Timing-driven Global Placement with Momentum-based Net Weighting](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Mar 14-23, 2022.
- [C87] Haoyu Yang, Kit Fung, Yuxuan Zhao, **Yibo Lin** and Bei Yu\*, “[Mixed-Cell-Height Legalization on CPU-GPU Heterogeneous Systems](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Mar 14-23, 2022.

- [C86] Xun Jiang, **Yibo Lin** and Zhongfeng Wang\*, “[FPGA-Accelerated Maze Routing Kernel for VLSI Designs](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC), Virtual Conference, Jan 17-20, 2022.
- [C85] Kexing Zhou, Zizheng Guo, Tsung-Wei Huang and **Yibo Lin**\*, “[Efficient Critical Paths Search Algorithm using Mergeable Heap](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC), Virtual Conference, Jan 17-20, 2022.
- [J84] Yibai Meng, Wuxi Li, **Yibo Lin**\* and David Z. Pan, “[elfPlace: Electrostatics-based Placement for Large-Scale Heterogeneous FPGAs](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jan, 2022.
- [C83] Zizheng Guo, Tsung-Wei Huang and **Yibo Lin**\*, “[A Provably Good and Practically Efficient Algorithm for Common Path Pessimism Removal in Large Designs](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.
- [C82] Guannan Guo, Tsung-Wei Huang\*, **Yibo Lin** and Martin Wong, “[GPU-accelerated Path-based Timing Analysis](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.
- [C81] Zizheng Guo, Jing Mai and **Yibo Lin**\*, “[Ultrafast CPU/GPU Kernels for Density Accumulation in Placement](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.
- [C80] Xiaohan Gao, Mingjie Liu, David Z. Pan and **Yibo Lin**\*, “[Interactive Analog Layout Editing with Instant Placement Legalization](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.
- [J79] **Yibo Lin**\*, “[模拟电路版图自动化与智能设计](#)”, 中国计算机学会通讯, Dec, 2021.
- [J78] Wei Li, Yuzhe Ma, Qi Sun, Zhang Lu, **Yibo Lin**, Iris Hui-Ru Jiang, Bei Yu\* and David Z. Pan, “[OpenMPL: An Open Source Layout Decomposer](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Nov, 2021.
- [C77] Zizheng Guo, Tsung-Wei Huang and **Yibo Lin**\*, “[HeteroCPR: Accelerating Common Path Pessimism Removal with Heterogeneous CPU-GPU Parallelism](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Virtual Conference, Nov 01-04, 2021.
- [C76] Guannan Guo, Tsung-Wei Huang\*, **Yibo Lin** and Martin Wong, “[GPU-accelerated Critical Path Generation with Path Constraints](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Virtual Conference, Nov 01-04, 2021.
- [C75] Tong Qu, **Yibo Lin**, Tianyang Gai, Xiaojing Su, Shuhan Wang, Bojie Ma, Yajuan Su\* and Yayi Wei\*, “[Litho-Aware Redundant Local-Loop Insertion Framework With Convolutional Neural Network](#)”, Proceedings of SPIE, San Jose, CA, Sep 27, 2021.
- [J74] Tsung-Wei Huang\*, **Yibo Lin**, Chun-Xun Lin, Guannan Guo and Martin Wong, “[Cpp-Taskflow: A General-purpose Parallel Task Programming System at Scale](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Aug, 2021.
- [J73] **Yibo Lin**\*, Zixuan Jiang, Jiaqi Gu, Wuxi Li, Shounak Dhar, Haoxing Ren, Brucek Khailany and David Z. Pan, “[DREAMPlace: Deep Learning Toolkit-Enabled GPU Acceleration for Modern VLSI Placement](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Apr, 2021. (**Best Paper Award**)

- [J72] Junzhe Cai, Changhao Yan\*, Yudong Tao, **Yibo Lin**, Sheng-Guo Wang, David Z. Pan and Xuan Zeng, “[A Novel and Unified Full-chip CMP Model Aware Dummy Fill Insertion Framework with SQP-Based Optimization Method](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Mar, 2021.
- [J71] Mohamed Baker Alawieh, **Yibo Lin**, Zaiwei Zhang, Meng Li, Qixing Huang and David Z. Pan\*, “[GAN-SRAF: Sub-Resolution Assist Feature Generation using Generative Adversarial Networks](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Feb, 2021.
- [J70] **Yibo Lin\***, Xiaohan Gao, Tinghuan Chen and Bei Yu\*, “[机器学习辅助数字集成电路后端设计方法](#)”, 微纳电子与智能制造, Feb, 2021.
- [J69] Cheng Zhuo\*, Zizheng Guo, Xiao Dong, Qing He and **Yibo Lin**, “[先进工艺下的数字签核](#)”, 微纳电子与智能制造, Feb, 2021.
- [C68] Tong Qu, **Yibo Lin**, Zongqing Lu, Yajuan Su\* and Yayi Wei\*, “[Asynchronous Reinforcement Learning Framework for Net Order Exploration in Detailed Routing](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Virtual Conference, Feb 01-05, 2021.
- [C67] Siting Liu, Qi Sun, Peiyu Liao, **Yibo Lin** and Bei Yu\*, “[Global Placement with Deep Learning-Enabled Explicit Routability Optimization](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Virtual Conference, Feb 01-05, 2021.
- [C66] Hongjia Li, Mengshu Sun, Tianyun Zhang, Olivia Chen, Nobuyuki Yoshikawa, Bei Yu, Yanzhi Wang\* and **Yibo Lin**, “[Towards AQFP-Capable Physical Design Automation](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Virtual Conference, Feb 01-05, 2021.
- [C65] Xiaohan Gao, Chenhui Deng, Mingjie Liu, Zhiru Zhang, David Z. Pan and **Yibo Lin\***, “[Layout Symmetry Annotation for Analog Circuits with Graph Neural Networks](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 18-21, 2021.
- [C64] **Yibo Lin\***, “[Deep Learning for Mask Synthesis and Verification: A Survey](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 18-21, 2021. (Invited Paper)
- [J63] **Yibo Lin\***, Wuxi Li, Jiaqi Gu, Haoxing Ren, Brucek Khailany and David Z. Pan, “[ABCD-Place: Accelerated Batch-based Concurrent Detailed Placement on Multi-threaded CPUs and GPUs](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Dec, 2020.
- [C62] Jiaqi Gu, Zixuan Jiang, **Yibo Lin** and David Z. Pan\*, “[DREAMPlace 3.0: Multi-Electrostatics Based Robust VLSI Placement with Region Constraints](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Nov 2-5, 2020.
- [C61] Zizheng Guo, Tsung-Wei Huang and **Yibo Lin\***, “[GPU-Accelerated Static Timing Analysis](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Nov 2-5, 2020.
- [C60] **Yibo Lin\***, “[GPU Acceleration in VLSI Back-end Design: Overview and Case Studies](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Nov 2-5, 2020. (Invited tutorial)

- [C59] Wei Ye, Mohamed Baker Alawieh, Yuki Watanabe, Shigeki Nojima, **Yibo Lin** and David Z. Pan\*, “[TEMPO: Fast Mask Topography Effect Modeling with Deep Learning](#)”, ACM International Symposium on Physical Design (ISPD), Taipei, Taiwan, Sep 20-23, 2020. (**Best Paper Award**)
- [J58] Jing Chen, Mohamed Baker Alawieh, **Yibo Lin**, Maolin Zhang, Jun Zhang, Yufeng Guo\* and David Z. Pan, “[Automatic Selection of Structure Parameters of Silicon on Insulator Lateral Power Device Using Bayesian Optimization](#)”, IEEE Electron Device Letters (EDL), Sep, 2020.
- [C57] Wei Li, Jialu Xia, Yuzhe Ma, Jialu Li, **Yibo Lin** and Bei Yu\*, “[Adaptive Layout Decomposition with Graph Embedding Neural Networks](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, Jul 19-23, 2020.
- [J56] Ying Chen, **Yibo Lin**, Rui Chen, Lisong Dong, Ruixuan Wu, Tianyang Gai, Le Ma, Yajuan Su\* and Yayi Wei\*, “[EUV Multilayer Defect Characterization via Cycle-Consistent Learning](#)”, Optics Express, Jun, 2020.
- [C55] **Yibo Lin**\*, David Z. Pan, Haoxing Ren and Brucek Khailany, “[DREAMPlace 2.0: Open-Source GPU-Accelerated Global and Detailed Placement for Large-Scale VLSI Designs](#)”, China Semiconductor Technology International Conference (CSTIC), Shanghai, China, Jun, 2020. (**Invited Paper**)
- [J54] Jing Chen, Mohamed Baker Alawieh, **Yibo Lin**, Maolin Zhang, Jun Zhang, Yufeng Guo\* and David Z. Pan, “[Powernet: SOI Lateral Power Device Breakdown Prediction With Deep Neural Networks](#)”, IEEE Access, Feb, 2020.
- [J53] Ying Chen, **Yibo Lin**, Lisong Dong, Tianyang Gai, Rui Chen, Yajuan Su\*, Yayi Wei\* and David Z. Pan, “[SoulNet: Ultrafast Optical Source Optimization Utilizing Generative Neural Networks for Advanced Lithography](#)”, Journal of Micro/Nanolithography, MEMS, and MOEMS (JM3), Nov, 2019.

===== Below are publications during Ph.D. and Postdoc =====

## Book Chapters

- [B1] **Yibo Lin** and David Z. Pan, “[Machine Learning in Physical Verification, Mask Synthesis, and Physical Design](#)”, Machine Learning in VLSI Computer-Aided Design, Springer, 2018, edited by Abe Elfedel, Duane Boning and Xin Li. (**Invited Book Chapter**)

## Conference and Journal Papers (\* denotes corresponding authors)

*Summary: DAC (7), ICCAD (7), IEEE TCAD (10), DATE (1), etc.*

- [J52] Hao Chen, Mingjie Liu, Biying Xu, Keren Zhu, Xiyuan Tang, Shaolan Li, **Yibo Lin**, Nan Sun and David Z. Pan\*, “[MAGICAL: An Open-Source Fully Automated Analog IC Layout System from Netlist to GDSII](#)”, IEEE Design & Test, Apr, 2021.
- [C51] Rachel Selina Rajarathnam, **Yibo Lin**, Yier Jin and David Z. Pan\*, “[ReGDS: A Reverse Engineering Framework from GDSII to Gate-level Netlist](#)”, IEEE International Workshop on Hardware-Oriented Security and Trust (HOST), San Jose, CA, May 4, 2020.
- [C50] Mingjie Liu, Wuxi Li, Keren Zhu, Biying Xu, **Yibo Lin**, Linxiao Shen, Xiyuan Tang, Nan Sun and David Z. Pan\*, “[S3DET: Detecting System Symmetry Constraints for Analog Circuits with Graph Similarity](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Jan 13-16, 2020. (**Best Paper Nomination**)



- [C49] Mohamed Baker Alawieh, Wuxi Li, **Yibo Lin**, Love Singhal, Mahesh Iyer and David Z. Pan\*, “[High-Definition Routing Congestion Prediction for Large-Scale FPGAs](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Jan 13-16, 2020.
- [C48] Wuxi Li, **Yibo Lin** and David Z. Pan\*, “[elfPlace: Electrostatics-based Placement for Large-Scale Heterogeneous FPGAs](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Westminster, CO, Nov 4-7, 2019.
- [C47] Keren Zhu, Mingjie Liu, **Yibo Lin**, Biying Xu, Shaolan Li, Xiyuan Tang, Nan Sun and David Z. Pan\*, “[GeniusRoute: A New Analog Routing Paradigm Using Generative Neural Network Guidance](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Westminster, CO, Nov 4-7, 2019.
- [C46] Chengyue Gong, Zixuan Jiang, Dilin Wang, **Yibo Lin**, Qiang Liu and David Z. Pan\*, “[Mixed Precision Neural Architecture Search for Energy Efficient Deep Learning](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Westminster, CO, Nov 4-7, 2019.
- [C45] Biying Xu, Keren Zhu, Mingjie Liu, **Yibo Lin**, Shaolan Li, Xiyuan Tang, Nan Sun and David Z. Pan\*, “[MAGICAL: Toward Fully Automated Analog IC Layout Leveraging Human and Machine Intelligence](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Westminster, CO, Nov 4-7, 2019. (**Invited Paper**)
- [J44] **Yibo Lin**, Meng Li, Yuki Watanabe, Taiki Kimura, Tetsuaki Matsunawa, Shigeki Nojima and David Z. Pan\*, “[Data Efficient Lithography Modeling with Transfer Learning and Active Data Selection](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Oct, 2019.
- [C43] Wei Li, Yuzhe Ma, Qi Sun, **Yibo Lin**, Iris Hui-Ru Jiang, Bei Yu and David Z. Pan, “[Open-MPL: An Open Source Layout Decomposer](#)”, International Conference on ASIC (ASICON), Chongqing, China, Oct, 2019. (**Invited Paper**)
- [C42] **Yibo Lin**, Shounak Dhar, Wuxi Li, Haoxing Ren, Brucek Khailany and David Z. Pan\*, “[DREAMPlace: Deep Learning Toolkit-Enabled GPU Acceleration for Modern VLSI Placement](#)”, ACM/IEEE Design Automation Conference (DAC), Las Vegas, NV, Jun 2-6, 2019. (**Best Paper Award**)
- [C41] Wei Ye, Mohamed Baker Alawieh, **Yibo Lin** and David Z. Pan\*, “[LithoGAN: End-to-End Lithography Modeling with Generative Adversarial Networks](#)”, ACM/IEEE Design Automation Conference (DAC), Las Vegas, NV, Jun 2-6, 2019. (**Best Paper Nomination**)
- [C40] Biying Xu, **Yibo Lin**, Xiyuan Tang, Shaolan Li, Linxiao Shen, Nan Sun and David Z. Pan\*, “[WellGAN: Generative-Adversarial-Network-Guided Well Generation for Analog/Mixed-Signal Circuit Layout](#)”, ACM/IEEE Design Automation Conference (DAC), Las Vegas, NV, Jun 2-6, 2019.
- [C39] Mohamed Baker Alawieh, **Yibo Lin**, Zaiwei Zhang, Meng Li, Qixing Huang and David Z. Pan\*, “[GAN-SRAF: Sub-Resolution Assist Feature Generation Using Conditional Generative Adversarial Networks](#)”, ACM/IEEE Design Automation Conference (DAC), Las Vegas, NV, Jun 2-6, 2019.
- [C38] **Yibo Lin**, Zhao Song and Lin F. Yang\*, “[Towards a Theoretical Understanding of Hashing-Based Neural Nets](#)”, International Conference on Artificial Intelligence and Statistics (AIS-



TATS), Okinawa, Japan, Apr 16-18, 2019.

- [C37] Biying Xu, Shaolan Li, Chak-Wa Pui, Derong Liu, Linxiao Shen, **Yibo Lin**, Nan Sun and David Z. Pan\*, “[Device Layer-Aware Analytical Placement for Analog Circuits](#)”, ACM International Symposium on Physical Design (ISPD), San Francisco, CA, Apr 14-17, 2019. (**Best Paper Nomination**)
- [J36] Ying Chen, **Yibo Lin**, Tianyang Gai, Yajuan Su\*, Yayi Wei\* and David Z. Pan, “[Semi-Supervised Hotspot Detection with Self-Paced Multi-Task Learning](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Apr, 2019.
- [C35] Wei Ye, Mohamed Baker Alawieh, Meng Li, **Yibo Lin** and David Z. Pan\*, “[Litho-GPA: Gaussian Process Assurance for Lithography Hotspot Detection](#)”, IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Florence, Italy, Mar 25-29, 2019.
- [J34] Jing Chen, **Yibo Lin**, Yufeng Guo, Maolin Zhang, Mohamed Baker Alawieh and David Z. Pan\*, “[Lithography Hotspot Detection Using a Double Inception Module Architecture](#)”, Journal of Micro/Nanolithography, MEMS, and MOEMS (JM3), Mar, 2019.
- [C33] Ying Chen, **Yibo Lin**, Tianyang Gai, Yajuan Su\*, Yayi Wei\* and David Z. Pan, “[Semi-Supervised Hotspot Detection with Self-Paced Multi-Task Learning](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 21-24, 2019.
- [C32] Wei Ye, Mohamed Baker Alawieh, **Yibo Lin** and David Z. Pan\*, “[Tackling Signal Electromigration with Learning-Based Detection and Multistage Mitigation](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 21-24, 2019.
- [C31] Wei Ye, **Yibo Lin**, Meng Li, Qiang Liu and David Z. Pan\*, “[LithoROC: Lithography Hotspot Detection with Explicit ROC Optimization](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 21-24, 2019. (**Invited Paper**)
- [C30] **Yibo Lin**, Mohamed Baker Alawieh, Wei Ye and David Z. Pan\*, “[Machine Learning for Yield Learning and Optimization](#)”, IEEE International Test Conference (ITC), Phoenix, Arizona, Oct, 2018. (**Invited Paper**)
- [J29] **Yibo Lin**, Bei Yu, Meng Li and David Z. Pan\*, “[Layout Synthesis for Topological Quantum Circuits with 1D and 2D Architectures](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Aug, 2018.
- [J28] Meng Li, Bei Yu, **Yibo Lin**, Xiaoqing Xu, Wuxi Li and David Z. Pan, “[A practical split manufacturing framework for trojan prevention via simultaneous wire lifting and cell insertion](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jul, 2018.
- [C27] Jiong Zhang, **Yibo Lin**, Zhao Song and Inderjit S Dhillon\*, “[Learning Long Term Dependencies via Fourier Recurrent Units](#)”, International Conference on Machine Learning (ICML), Stockholm, Sweden, Jun 10-15, 2018.
- [J26] Xiaoqing Xu, **Yibo Lin**, Meng Li, Tetsuaki Matsunawa, Shigeki Nojima, Chikaaki Kodama, Toshiya Kotani and David Z. Pan\*, “[Subresolution Assist Feature Generation With Supervised Data Learning](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2018.
- [J25] **Yibo Lin**, Bei Yu, Xiaoqing Xu, Jih-Rong Gao, Natarajan Viswanathan, Wen-Hao Liu,

- Zhuo Li, Charles J Alpert and David Z. Pan\*, “[MrDP: Multiple-row detailed placement of heterogeneous-sized cells for advanced nodes](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2018.
- [J24] Wuxi Li, **Yibo Lin**, Meng Li, Shounak Dhar and David Z. Pan\*, “[UTPlaceF 2.0: A High-Performance Clock-Aware FPGA Placement Engine](#)”, ACM Transactions on Design Automation of Electronic Systems (TODAES), Jun, 2018.
- [C23] **Yibo Lin**, Yuki Watanabe, Taiki Kimura, Tetsuaki Matsunawa, Shigeki Nojima, Meng Li and David Z. Pan\*, “[Data Efficient Lithography Modeling with Residual Neural Networks and Transfer Learning](#)”, ACM International Symposium on Physical Design (ISPD), Monterey, CA, Mar 25-28, 2018.
- [C22] Meng Li, Bei Yu, **Yibo Lin**, Xiaoqing Xu, Wuxi Li and David Z. Pan\*, “[A Practical Split Manufacturing Framework for Trojan Prevention via Simultaneous Wire Lifting and Cell Insertion](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Jeju, Korea, Jan 22-25, 2018.
- [C21] Che-Lun Hsu, Shaofeng Guo, **Yibo Lin**, Xiaoqing Xu, Meng Li, Runsheng Wang, Ru Huang and David Z Pan, “[Layout-dependent aging mitigation for critical path timing](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Jeju, Korea, Jan 22-25, 2018.
- [C20] **Yibo Lin**, Peter Debacker, Darko Trivkovic, Ryoung-han Kim, Praveen Raghavan and David Z. Pan\*, “[Patterning Aware Design Optimization of Selective Etching in N5 and Beyond](#)”, IEEE International Conference on Computer Design (ICCD), Boston, MA, Nov 5-8, 2017.
- [C19] **Yibo Lin**, Xiaoqing Xu, Jiaojiao Ou and David Z Pan, “[Machine learning for mask/wafer hotspot detection and mask synthesis](#)”, Photomask Technology, Oct 16, 2017. (**Invited Paper**)
- [J18] **Yibo Lin**, Bei Yu and David Z. Pan\*, “[High performance dummy fill insertion with coupling and uniformity constraints](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Sep, 2017.
- [C17] Wei Ye, **Yibo Lin**, Xiaoqing Xu, Wuxi Li, Yiwei Fu, Yongsheng Sun, Canhui Zhan and David Z. Pan\*, “[Placement Mitigation Techniques for Power Grid Electromigration](#)”, IEEE International Symposium on Low Power Electronics and Design (ISLPED), Taipei, Jul 24-26, 2017.
- [J16] **Yibo Lin**, Bei Yu, Biying Xu and David Z. Pan\*, “[Triple patterning aware detailed placement toward zero cross-row middle-of-line conflict](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jul, 2017.
- [J15] Xiaoqing Xu, **Yibo Lin**, Meng Li, Jiaojiao Ou, Brian Cline and David Z. Pan\*, “[Redundant local-Loop insertion for unidirectional routing](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jul, 2017.
- [C14] Xiaoqing Xu, **Yibo Lin**, Vinicius Livramento and David Z. Pan\*, “[Concurrent Pin Access Optimization for Unidirectional Routing](#)”, ACM/IEEE Design Automation Conference (DAC), Austin, TX, Jun 18-22, 2017.
- [J13] **Yibo Lin**, Bei Yu, Yi Zou, Zhuo Li, Charles J Alpert and David Z. Pan\*, “[Stitch aware detailed placement for multiple e-beam lithography](#)”, Integration, the VLSI Journal, Jun,

2017. (**Best Paper Award**)

- [J12] **Yibo Lin**, Xiaoqing Xu, Bei Yu, Ross Baldick and David Z. Pan\*, “[Triple/quadruple patterning layout decomposition via linear programming and iterative rounding](#)”, Journal of Micro/Nanolithography, MEMS, and MOEMS (JM3), Jun, 2017.
- [C11] Jiaojiao Ou, Bei Yu, Xiaoqing Xu, Joydeep Mitra, **Yibo Lin** and David Z. Pan\*, “[DSAR: DSA aware routing with simultaneous DSA guiding pattern and double patterning assignment](#)”, ACM International Symposium on Physical Design (ISPD), Portland, OR, Mar 19-22, 2017.
- [C10] **Yibo Lin**, Bei Yu, Xiaoqing Xu, Jhih-Rong Gao, Natarajan Viswanathan, Wen-Hao Liu, Zhuo Li, Charles J Alpert and David Z. Pan\*, “[MrDP: Multiple-row detailed placement of heterogeneous-sized cells for advanced nodes](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Austin, TX, Nov 7-10, 2016.
- [C9] Yudong Tao, Changhao Yan\*, **Yibo Lin**, Sheng-Guo Wang, David Z. Pan and Xuan Zeng, “[A novel unified dummy fill insertion framework with SQP-based optimization method](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Austin, TX, Nov 7-10, 2016.
- [C8] **Yibo Lin**, Bei Yu and David Z. Pan\*, “[Detailed placement in advanced technology nodes: a survey](#)”, IEEE International Conference on Solid-State and Integrated Circuit Technology (ICSICT), Hangzhou, China, Oct 25-28, 2016. (**Invited Paper**)
- [J7] Bei Yu, Xiaoqing Xu, Subhendu Roy, **Yibo Lin**, Jiaojiao Ou and David Z. Pan\*, “[Design for manufacturability and reliability in extreme-scaling VLSI](#)”, Science China Information Sciences, May, 2016. (**Invited Paper**)
- [C6] **Yibo Lin**, Xiaoqing Xu, Bei Yu, Ross Baldick and David Z. Pan\*, “[Triple/quadruple patterning layout decomposition via novel linear programming and iterative rounding](#)”, Proceedings of SPIE, San Jose, CA, Feb 21-25, 2016. (**Best Student Paper Award**)
- [C5] **Yibo Lin**, Bei Yu, Yi Zou, Zhuo Li, Charles J Alpert and David Z. Pan\*, “[Stitch aware detailed placement for multiple e-beam lithography](#)”, IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Macau, China, Jan 25-28, 2016.
- [C4] **Yibo Lin**, Bei Yu, Biying Xu and David Z. Pan\*, “[Triple patterning aware detailed placement toward zero cross-row middle-of-line conflict](#)”, IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Austin, TX, Nov 2-6, 2015.
- [C3] **Yibo Lin**, Bei Yu and David Z. Pan\*, “[High performance dummy fill insertion with coupling and uniformity constraints](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 7-11, 2015.
- [C2] David Z. Pan\*, Lars Liebmann, Bei Yu, Xiaoqing Xu and **Yibo Lin**, “[Pushing multiple patterning in sub-10nm: are we ready?](#)”, ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 7-11, 2015. (**Invited Paper**)
- [J1] Bei Yu, Xiaoqing Xu, Jhih-Rong Gao, **Yibo Lin**, Zhuo Li, Charles Alpert and David Z. Pan\*, “[Methodology for standard cell compliance and detailed placement for triple patterning lithography](#)”, IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), May, 2015.